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Route to a direct-gap silicon allotrope Si₃₂

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Abstract

Using swarm-intelligence-based structure prediction methods, we predict a novel direct bandgap silicon allotrope with open channels at ambient conditions. This silicon phase, termed Si₃₂, can be produced by removing Sr atoms from a new *Cmcm*-SrSi₈ clathrate-like compound, which is calculated to be thermodynamically stable under epitaxial strain at high pressures. Si₃₂ is predicted to have a direct bandgap of ~ 1.15 eV and exceptional optical properties. The prediction of novel silicon clathrate-like structure paves the way for the exploration of novel silicon phases with extensive application possibilities.

Keywords: structure prediction, first-principle calculation, silicon clathrate, photo-voltaic cell

 Supplementary material for this article is available [online](#)

(Some figures may appear in colour only in the online journal)

1. Introduction

Over the past decades, silicon has been the primary building block of contemporary technology due to its inherent benefits, including vast abundance, high stability, and the ability for doping by other elements. In addition to its numerous applications in the microelectronics sector, silicon also plays a significant role in solar energy generation. At the moment, the vast majority of solar cells are built of the diamond Si (*d*-Si), which is the most thermodynamically stable form of silicon at ambient conditions. While *d*-Si dominates the photovoltaic market owing to its various advantages, it has considerable limitations that prevent *d*-Si from being an optimum absorber of sunlight [1–4]. The bandgap of *d*-Si is indirect and its optical gap is larger than 3 eV, which means phonons are requisite to mediate electronic excitations in the optical regime for *d*-Si [1]. This limits the use of *d*-Si in thin-film solar cells or other high-performance platforms for optical applications, which are crucial for developing a clean and efficient method of converting solar energy to electricity and therefore reducing our reliance on fossil fuels. Driven by a strong desire for clean energy, much effort has been expanded to find new Si allotropes with absorption spectra that strongly overlap with the solar spectrum [5–13].

Experimental studies of Si allotropes have revealed a very rich phase space with a sequence of structures that appear in the high-pressure regime [14, 15], and several of these structures are recoverable to ambient conditions, allowing them to be considered as potential solar devices [16–18]. However, when it comes to photovoltaic applications, there is no known silicon phase that meets the 1.15 or 1.34 eV direct bandgap criterion for the maximum solar converting efficiency [19–21]. For example, by decompressing the high-pressure phase Si-II, the metastable Si phases R8 and BC8 could be quenched to ambient pressure, but none of them shows an enhanced photon absorption property within the visible light range compared to diamond Si [17, 18, 22, 23]. Extensive *ab initio* studies of structural and electronic properties of type-I (Si₄₆) and type-II (Si₃₄, often referred to as Si₁₃₆) silicon clathrates have shown that direct optical transitions are forbidden by symmetry in a dipole approximation [24, 25]. The crystalline 4H hexagonal silicon (4H-Si), which has only recently been synthesized through a metastable phase transformation, has an indirect gap of ~ 1.2 eV [26].

On the other hand, it was predicted that many silicon allotropes with direct or quasidirect bandgaps could boost light absorption significantly, but very few syntheses have been recorded. In silicon, the tetrahedral bonding creates an extremely complicated energy landscape, with multiple allotropes with very minor variations in energy from the ground state [7, 8, 10, 27–29]. As a result, access to these

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hypothetical metastable phases is extremely difficult in concept. Another method for creating new material could be the use of precursors. For example, N-doped graphene nanoribbon, which is difficult to make directly, can be fabricated using various precursors [30]. Type-II silicon clathrate and germanium clathrate were previously synthesized using silicon-rich substances, both of which employed chemical precursors produced at atmospheric pressure [31, 32].

Pressure also has long been an effective technique for discovering materials that are otherwise inaccessible at ambient conditions. Thus, Kim *et al* recently reported the successful synthesis of a novel allotrope of silicon (Si_{24}), using a high-pressure precursor NaSi_6 , in a unique two-step methodology [33]. Si_{24} has a quasidirect bandgap, which has the potential to significantly enhance optoelectronic characteristics when compared to *d*-Si. However, Si_{24} is still not a direct bandgap phase, which means the direct bandgap does not occur at the high symmetry point, although the indirect and direct gaps are almost degenerate. This characteristic could hinder the capability for light absorption efficiency that may be achieved by the material.

Along with hydrostatic pressure, it has been shown that nonhydrostatic pressure also promotes the synthesis of new materials. For example, it was discovered that a large uniaxial stress fosters the formation of cubane-derived nanowires and an epitaxial strain can be used to stabilize ferroelectric oxynitrides [34, 35]. In this work, we propose a strategy for designing a synthesizable direct bandgap Si phase (Si_{32}) through a high-pressure precursor SrSi_8 that is thermodynamically stable under epitaxial strain. Similar to NaSi_6 , SrSi_8 also contains a channel-like sp^3 silicon network, and thus Sr atoms can be removed by a similar approach of thermal degassing process.

2. Method

Global structural optimization was performed using the calypso code [36–38] with the particle swarm optimization algorithm, which has successfully predicted structures of various systems ranging from elements to binary and ternary compounds [39–43]. The energetic calculations were carried out using density functional theory within the Perdew–Burke–Ernzerhof (PBE) generalized gradient approximation [44, 45] as implemented in the Vienna *ab initio* simulation package (vasp) [46]. We used projector augmented waves with $4s^2 4p^6 5s^2$ and $3s^2 3p^2$ electrons as valence for Sr and Si atoms, respectively. The electronic wave functions were expanded in a plane-wave basis set with a cutoff energy of 400 eV. Monkhorst–Pack *k*-point meshes [47] with a grid of spacing $0.04 \times 2\pi \text{\AA}^{-1}$ for Brillouin zone sampling were chosen. To determine the dynamical stability of the studied structures, we performed phonon calculations by using the finite displacement approach, as implemented in the phonopy code [48]. The hybrid Heyd–Scuseria–Ernzerhof functional (HSE06) [49] was used to calculate the accurate bandgap of the new silicon allotrope. Absorption spectra were calculated by determined by solving the Bethe–Salpeter equation (BSE)

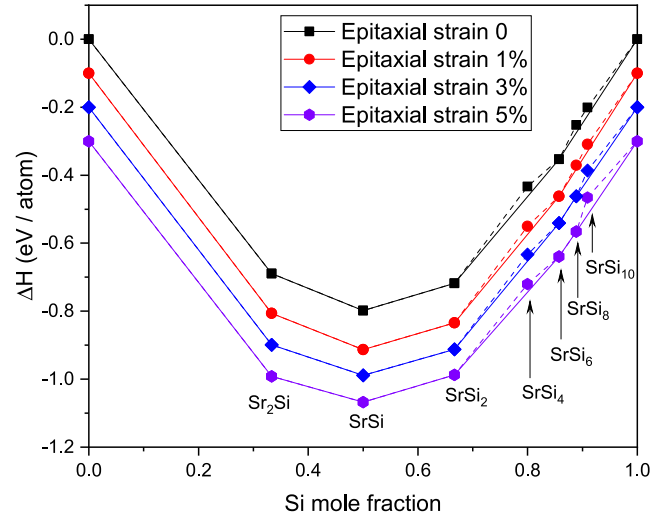


Figure 1. Predicted formation enthalpies of various Sr–Si compounds at 10 GPa with respect to elemental decomposition into Sr and Si as a function of epitaxial strain. Dashed lines connect data points, and solid lines denote the convex hull. Compounds with enthalpy or free-energy data located on the convex hull are stable against decomposition. For clarity, we have offset the enthalpy plots by -0.1 , -0.2 , and -0.3 eV at 1%, 3% and 5% epitaxial strain, respectively.

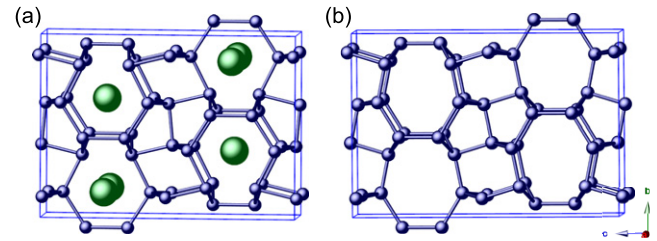


Figure 2. Top views of the crystal structures of (a) *Cmcmm*- SrSi_8 and (b) Si_{32} . Green and blue spheres represent Sr and Si atoms, respectively. The rectangles denote the unit cells.

on the top of G_0W_0 approximation [50, 51] within the vasp code [46].

3. Results

Calculations of structure prediction were carried out at a pressure of 10 GPa using up to four formula units per simulation cell. We compute the enthalpy of the crystal phases discovered by the structure search procedure at various Sr:Si ratios to select the most viable structure at each composition. We use the following equation to calculate the formation enthalpy (per atom) for composition Sr_xSi_y :

$$\Delta H(\text{Sr}_x\text{Si}_y) = [H(\text{Sr}_x\text{Si}_y) - xH(\text{Sr}) - yH(\text{Si})] / (x + y),$$

where $\Delta H(\text{Sr}_x\text{Si}_y)$ is the formation enthalpy per atom of a compound with this stoichiometry. We next plot the convex hull, as illustrated in figure 1, revealing stable products of Sr and Si reactions. This investigation resulted in the identification of several Sr–Si compounds that are stable against decomposition at a pressure of 10 GPa, including Sr_2Si , SrSi , SrSi_2 ,

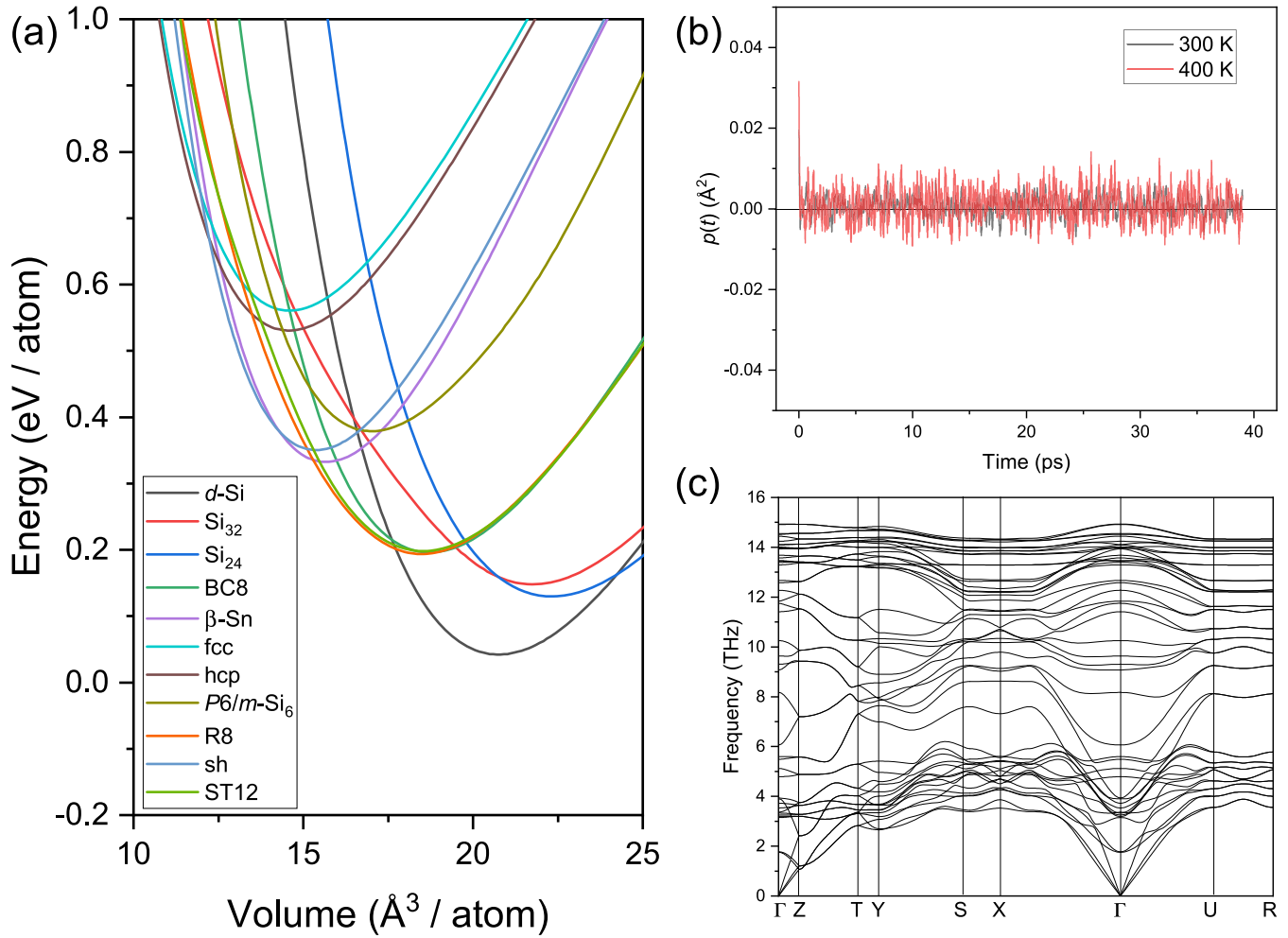


Figure 3. (a) Energy vs volume curves of various Si phases. (b) The plots of position correlation functions, $p(t)$, for Si_{32} . The position correlation function for a chosen atom, i , is given by $p(t) = \langle (\mathbf{r}_i(t + t_0) - \mathbf{R}_i^0)(\mathbf{r}_i(t_0) - \mathbf{R}_i^0) \rangle$, where $\mathbf{r}_i$ is the time-varying position of the atom and $\mathbf{R}_i^0$ is the position of that atom's lattice site in the perfect starting structure. The angular brackets denote the thermal average, which in practice is evaluated as an average over time origins, t_0 , and atoms i . For long times t , vibrational displacements become uncorrelated, so that $p(t) \rightarrow \langle \mathbf{r}_i - \mathbf{R}_i^0 \rangle^2$, and if all atoms vibrate about the starting lattice sites, $\langle \mathbf{r}_i - \mathbf{R}_i^0 \rangle^2 = 0$, so that $p(t) \rightarrow 0$ as $t \rightarrow \infty$. (c) Phonon dispersion relations of Si_{32} at ambient pressure.

and SrSi_6 . In the Si-rich region, the most stable phase is the *Cmcm* structure of SrSi_6 , which has been discovered via high-pressure synthesis [52]. SrSi_6 shares the same crystal structure as NaSi_6 and thus can also be used as a precursor to produce the Si_{24} framework [33]. In addition, we find that all the metastable configurations are low-lying structures and are very close to the convex hull, though not on it. For example, SrSi_8 is only 20 meV atom^{-1} above the convex hull, which is in the metastability range and makes it possible to synthesize the structure under high-pressure conditions [53]. Given that non-hydrostatic pressure has been proven to significantly improve the stability or properties of a variety of materials [34, 35], we next evaluate the stabilities of the predicted phases under epitaxial strain. For the *Cmcm* structure of SrSi_8 , we fix the lattice parameters of b and c with 1% to 5% strains, and relax a axis and atomic positions. As shown in figure 1, SrSi_8 becomes thermodynamically stable when the epitaxial strain reaches 3%, whereas SrSi_4 and SrSi_{10} remain metastable (dynamically stable under the epitaxial strain, figure S1

(<https://stacks.iop.org/JPCM/34/154006/mmedia>)). The dynamic stabilities of the *Cmcm*- SrSi_8 were examined by calculating the phonon spectra, and no imaginary phonon frequencies were found in the whole Brillouin zone under the epitaxial strain of 0%–5% at 10 GPa (figure S2). The stability of the strained *Cmcm*- SrSi_8 is encouraging for future synthesis efforts through strain engineering.

The crystal structure of *Cmcm*- SrSi_8 is comprised of a three-dimensional network of four-bonded Si atoms (figure 2(a)). The Si framework is constructed of Si channels enclosing strontium atoms that are formed by puckered eight-membered rings along the a axis. The channels are connected by the sections constructed by five- and six-membered rings, in contrast to the *Cmcm*- NaSi_6 , where the channels are joined via solely five-membered rings along the c -axis [33]. Phonon spectra calculations indicate that the *Cmcm* structure of SrSi_8 is dynamically stable at ambient pressure (figure S3), implying that *Cmcm*- SrSi_8 is recoverable to the ambient conditions. The calculated electron localization function reveals strong

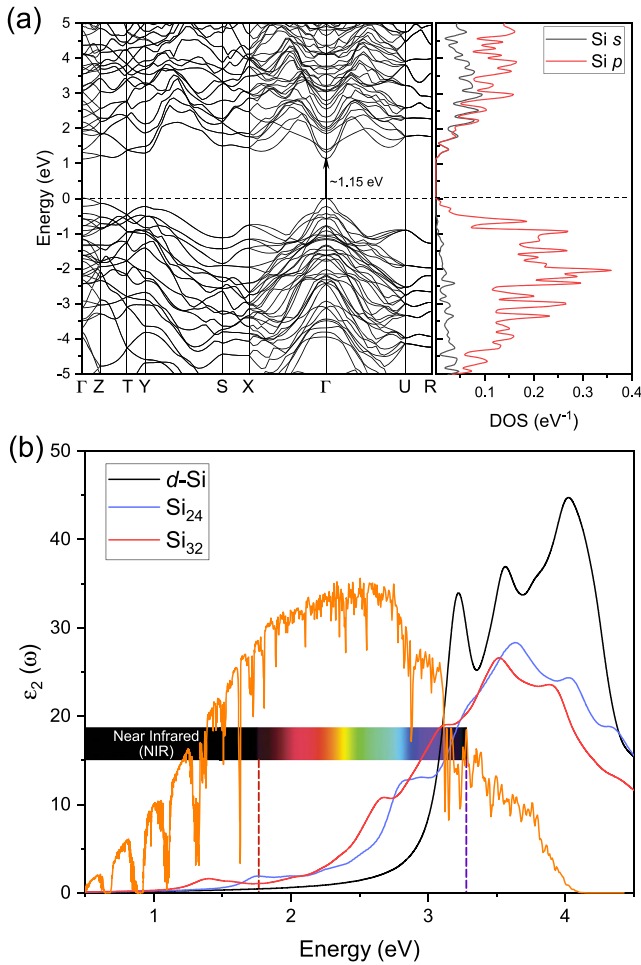


Figure 4. (a) Calculated band structures and electronic densities of state of Si_{32} using the HSE hybrid functional at ambient pressure with the Fermi level set to zero. (b) Calculated BSE absorption (ϵ_2) for Si_{32} (green), Si_{24} (blue) and $d\text{-Si}$ (black) compared with the reference AM 1.5 solar spectral irradiance (orange).

covalent bonding between sp^3 -hybridized Si atoms and only weak interactions with the Sr^{2+} guest (figure S4). This weakly bound nature suggests that the Sr atoms can be removed from the open channels using a thermal degassing process after pressure is released from Cmcm-SrSi_8 , and one can subsequently get a new Si allotrope (Si_{32} , 32 atoms per unit cell, figure 2(b)) in the space group Cmcm .

Phonon computations were also carried out to determine the dynamical lattice stability of the Si_{32} framework. There were no observed imaginary phonon modes, verifying the structural stability at atmospheric pressure (figure 3(c)). Figure 3(a) plots the relative energies with respect to $d\text{-Si}$ as a function of the atomic volume for several well-known Si phases. The total energy of Si_{32} is estimated to be approximately $110 \text{ meV atom}^{-1}$ higher than that of $d\text{-Si}$. It is also required to determine whether the Si_{32} phase is thermally stable for use in room-temperature solar cells. In order to investigate this possibility, we performed first-principles NPT molecular dynamic simulations using a large supercell with 192 atoms with temperature and pressure controlled via the Langevin thermostat [54] and Parrinello–Rahman barostat [55, 56],

respectively. Each simulation was performed for 40 ps with a 1 fs time step. Figure S5 shows the fluctuations of the total potential energy as a function of simulation time. After 40 ps, we find no structure degradation of the Si_{32} structure at 300 and 400 K under ambient pressure, save for modest thermal fluctuations. We plot the position correlation functions for Si_{32} at 300 and 400 K in figure 3(b), which also reveal that the Si_{32} structure is dynamically stable up to a temperature of at least 400 K.

In order to obtain further insights into the properties of Si_{32} , we first compute the electronic band structure using the PBE functional at ambient pressure. We find that Si_{32} has a direct bandgap of $\sim 0.48 \text{ eV}$ at the Γ point (figure S6), as opposed to the previously synthesized Si_{24} , which is reported to have an indirect bandgap [33]. It is well known that the PBE functional has a drawback of underestimating the bandgap of silicon. Therefore, we used the HSE hybrid functional to get precise bandgap estimations. Figure 4(a) shows the electronic band structure calculated using the HSE hybrid functional, and we can discover that the corrected direct bandgap is around 1.15 eV at the Γ point. The projected density of states reveals the sp^3 -hybridized bonding nature of the Si atoms and also confirms the bandgap (figure 4(a)). It is reported that the efficiency curve for light to electric power conversion is not completely smooth and contains two local maxima at 1.15 eV and 1.34 eV [20, 21]. As a result, the direct bandgap of $\sim 1.15 \text{ eV}$ is an ideal value for optimal light to electric power conversion efficiency [20, 21], and hence Si_{32} may have improved the photon absorption rate.

To evaluate possible improvements in light absorption capabilities, we calculated absorption spectra (imaginary part of dielectric function) of our predicted Si_{32} by solving the BSE [50, 57] and plotted it along with the reference air mass (AM) 1.5 solar spectral irradiance [58] (figure 4(b)). Additionally, we provide the imaginary component of the dielectric function for $d\text{-Si}$ and the previously synthesized Si_{24} for comparison, together with the spectrum of visible and near-infrared light [59]. Si_{32} exhibits optical absorption beginning at the 1.15 eV direct gap transition energy. We can find that the optical absorption of Si_{32} is greatly improved below 3 eV when compared to $d\text{-Si}$, and it is also better than Si_{24} in the visible light region [33]. Therefore, we propose that Si_{32} is an excellent contender for solar cell absorber material.

4. Conclusion

In conclusion, we predict a new Cmcm-SrSi_8 compound that can be produced under epitaxial strain conditions at high pressures using crystal structure prediction and first-principles total energy calculations. The stability of Cmcm-SrSi_8 down to zero pressure has been demonstrated using phonon spectrum calculations. After pressure release, we estimate that using the SrSi_8 clathrate-like structure as a precursor, the Sr degassing process will create a direct bandgap Si phase, designated Si_{32} . Due to the direct bandgap of 1.15 eV, we propose that Si_{32} could be a potential solar energy absorber. These encouraging results shed fresh light on the developmental approach for direct bandgap semiconductor design. Therefore, there is

potential to create a far more diverse landscape for silicon materials with advanced properties.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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